MIL-C-50269B (AR)
4 December 1987
SUPERSEDING
MIL-C-50269A (MU)
12 March 1971

MILITARY SPECIFICATION

CELLULOSE, WOODPULP (SULFATE) FOR NON-METALLIC CARTRIDGE CASES, KRAFT

This Specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of kraft process cellulose woodpulp for use in combustible ordnance items.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1320
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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturer in connection with specified acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of nongovernment documents which is current on the date of the solicitation.

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)
TEST METHODS

T-211 OM-85	Ash in Wood Pulp
T-223 HM-84	Pentosans in Wood and Pulp
T-230 OM-82	Viscosity of Pulp
T-233 HM-82	Fiber Length of Pulp by Classification
T-236 OS-76	Kappa Number of Pulp
T-429 HM-84	Alpha-Cellulose in Paper

(Application for copies should be made to the Technical Association of the Pulp and Paper Industry, Technology Park, P.O. Box 105113, Atlanta, GA, 30348).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E300	Recommended Practice for Sampling Industrial Chemicals
ASTM D-3951-82	Standard Practice for Commercial Packaging

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein,

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(except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The kraft process cellulose woodpulp shall comply with requirements given in this detailed specification as well as applicable drawings and referenced specifications.

3.2 Material. The material shall be kraft process cellulose woodpulp, meeting the requirements of this specification.

3.3 Chemical and physical properties. Chemical and physical properties shall be as indicated in Table I.

TABLE I - Chemical and physical properties

Property	Minimum	Maximum	Applicable Test Method
Fiber Length (mm)	2	3.5	4.3.1
Viscosity(centipoise)	10	25	4.3.2
Alpha Cellulose (%)	80		4.3.3
Pentosans (%)	5.0		4.3.4
Lignin (%)		1.5	4.3.5
Ash Content (%)		0.25	4.3.6

3.4 Workmanship. The best commercial practices shall be used in the manufacture of the product furnished under this specification. The woodpulp cellulose shall be free from dirt, grit, oil, grease and other foreign material when tested in accordance with 4.3.7.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements

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specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of any defective material.

4.2 Quality conformance inspection.

4.2.1 Lot formation. A lot shall consist of one or more batches of kraft process cellulose woodpulp produced by one manufacturer, at one plant, from the same materials under one set of operating conditions.

4.2.2 Inspection procedures. Inspection shall be made at the point of delivery, unless otherwise specified in the contract or order.

4.2.3 Sampling for chemical and physical properties. Two independent representative samples shall be withdrawn from each batch or lot of kraft process cellulose woodpulp in accordance with of ASTM E300. Each sample shall be tested independently in accordance with the methods in section 4.3. If either test sample fails to meet any of the requirements of section three of this specification, three additional representative samples shall be withdrawn from the failed batch/lot in accordance with ASTM E300. Each of the three samples shall be tested for all requirements in accordance with section 4.3 of this specification. If any of the three additional samples fails to meet any of the requirements of section three of this specification, the batch/lot shall be rejected.

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4.2.4 Sampling for packaging inspection. Sampling for packaging inspection shall be done in accordance with MIL-STD-105. An AQL of 0.40% for each major defect and an AQL of 0.65% for each minor defect shall be applied.

4.2.4.1 Woodpulp after packaging.

Catergories	Defect	Method of Inspection
Critical	None defined	
Major	None defined	
Minor		
201	Protective wrapping torn cut, perforated or incomplete	Visual
202	Marking misleading or unidentifiable	Visual

4.3 Methods of inspection. All tests given in this section shall be performed using prescribed procedures in the Technical Association of the Pulp and Paper Industry (TAPPI) Official Test Methods. See 6.3 for use of equivalent test methods.

4.3.1 Fiber length. The sample shall be tested for fiber length in accordance with TAPPI Test method TM-233 HM-82.

4.3.2 Viscosity. The sample shall be tested for viscosity in accordance with TAPPI Test method T-230 OM-82.

4.3.3 Alpha cellulose. The sample shall be tested for Alpha Cellulose in accordance with TAPPI Test method T429 HM-84.

4.3.4 Pentosans. The sample shall be tested for pentosans in accordance with TAPPI Test method T-223 HM-84.

4.3.5 Lignin. The sample shall be tested for lignin in accordance with TAPPI Test method T-236 OS-76.

4.3.6 Ash content. The sample shall be tested for ash content in accordance with TAPPI Test method T-211 OM-85.

4.3.7 Purity. The sample, when visually examined, shall be free from dirt, grit, oil, grease and other foreign material.

5. PACKAGING

5.1 Packing.

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5.1.1 Commercial. Packing shall be in accordance with the ASTM D-3951-82.

5.2 Marking.

5.2.1 Commercial. Marking shall be in accordance with ASTM D-3951-82 and MIL-STD-129.

6. NOTES

6.1 Intended use. Kraft process cellulose woodpulp is intended for use in the manufacture of combustible cartridge cases and other combustible ordnance products.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

- Title, number and date of this specification.

6.3 Equivalent test methods. The test methods given in this specification are the official methods to be used. The contractor may request using other methods providing that the proposed method is equivalent, (accuracy and precision) to the method given in this specification. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Technical Agency to: COMMANDER, ARDEC, ATTN: AMSMC-QAR-R(D), Picatinny Arsenal, NJ 07806-5000. This description should include, but not be limited to, the procedures used, the accuracy and precision, and drawings of any special equipment required.

6.4 Subject term (key word) listing.

Artillery
Combustible Components
Kraft
Woodpulp

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-50269B	CELLULOSE, WOODPULP (SULFATE)FOR NON-METALLIC CARTRIDGE CASE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
KRAFT	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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